Fidelity-Guided Multi-View Text Reconstruction from Exposed Overlapping RAG Embeddings

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Abstract. Some retrieval indexes use overlapping text chunks to preserve context at boundaries. If exposed, neighboring embeddings can be inverted independently, producing several noisy views of shared source content. We ask whether these views can be combined more effectively than treating each embedding alone.

We introduce *cross-view fidelity selection*. When two reconstructions cover the same content, it favors the reconstruction whose re-encoded text is more similar to the corresponding target embedding. The method reuses a score already computed by Vec2Text and needs neither a new inverter nor a learned confidence model. Because each score is measured against a different target, however, its usefulness across views is not guaranteed. We consider a post-compromise attacker who knows the chunk order and overlap geometry and has access to the target encoder and a matching frozen inverter, but not to the source text during assembly.

We conducted a post-hoc exploratory analysis with a matched GTR encoder and Vec2Text inverter. The evaluation used 60 overlap-stable Natural Questions spans held out for assembly evaluation. Fidelity-based selection chose the better reconstruction at 76 of 80 informative boundaries. This advantage was substantially weaker when the same scores were assigned to different views. On the same reconstructions, fidelity-based selection improved source-position recovery by 2.72 percentage points over always choosing the earlier chunk. An alignment-aware variant recovered slightly more content but also introduced more insertions and edits and produced fewer exact reconstructions. Thus, an existing inversion score helped combine correlated reconstructions in this matched setting; whether the effect transfers to other encoders, datasets, or inversion settings remains open.

Keywords: Embedding inversion · RAG · dense retrieval · privacy

1 Introduction

Repeated measurements can be more revealing than any measurement viewed alone. If two noisy observations describe partly the same object, their disagreement creates a choice: for the shared portion, one observation may preserve what the other loses. This principle matters for systems that store overlapping representations. Redundancy is introduced to preserve context at boundaries, yet the same redundancy can give an observer several imperfect views of the same underlying content.

Dense retrieval systems can create this situation. In retrieval-augmented generation (RAG), an indexer divides a corpus into passages and maps them to dense vectors; at inference time, a query vector retrieves nearby passages as nonparametric context for the generator [1,2,3]. This separation lets the generator use external knowledge without encoding every retrieved fact in its parameters. To avoid severing context at a passage boundary, an indexer may use overlapping windows. Under such a policy, disclosure of an ordered index exposes not only individual embeddings but also correlated representations of partly shared text.

For example, a 31-token window covering positions 0–30 and the next window covering 15–45 repeat positions 15–30. The two stored vectors encode different surrounding context but share 16 underlying tokens. After independent reconstruction, either view may preserve a shared token that the other substitutes or omits. Overlap therefore turns reconstruction from a collection of isolated decoding tasks into an assembly problem: the attacker must locate the shared region and decide which view to trust.

Embedding-inversion studies range from information-leakage analysis and generative sentence recovery to Vec2Text's sequence-level reconstruction [4,5,6]. Vec2Text frames inversion as controlled generation and iteratively re-embeds and corrects a hypothesis toward one target [6]. Under their respective model and access assumptions, these studies establish that dense representations can retain recoverable textual content. Their reconstructions remain noisy, however: they substitute, omit, or insert tokens, and two reconstructions of an overlap need not make the same error. This raises a question that single-embedding evaluation does not answer: when two independently inverted views cover the same source segment, can an attacker identify which reconstruction should own that segment without seeing the source?

Within one target's search, Vec2Text ranks sequence candidates by target–hypothesis cosine similarity [6]. It also reports a strong but imperfect empirical association between this score and reconstruction BLEU [6]. We call this available reconstruction-level signal *whole-view fidelity*. Different targets encode different nonshared context, so evidence for within-target ranking does not establish that raw scores are comparable across targets. We empirically test whether the unmodified scores can govern shared-content ownership across targets; we do not introduce a new inverter or confidence estimator.

Our contribution is *cross-view fidelity selection*, also called corresponding-fidelity ownership: assign an aligned shared segment to the reconstruction whose original view has higher whole-view fidelity. We separate this selection rule from the problem of locating corresponding reconstructed tokens. Fixed-coordinate fidelity (FC–F) applies the rule at relative positions implied by attacker-known chunk geometry; constrained progressive fidelity (CP–F) applies it after aligning reconstructed suffixes and prefixes to accommodate insertion and deletion drift. These are two realizations of the same selection principle. CP is useful when coordinates drift, but its additional alignment is not uniformly superior to FC. Both realizations use only reconstructed tokens, their scores, and known geom-

etry. We call this assembly *source-free*: the term describes assembly inputs, not the data used to train the inverter.

We evaluate this mechanism on 60 assembly-held-out, overlap-stable spans from Natural Questions (NQ) [7] with a matched GTR/Vec2Text pair. Fidelity selects the better shared segment at 76 of 80 informative boundaries, and reassigning the same scores weakens the advantage. FC–F recovers 222 more reference positions than lower-index ownership on the same views; CP–F adds 16 beyond FC–F but produces more insertions and edits and fewer span-exact matches. In this matched setting, the amount of source information recovered therefore depends not only on isolated inversion quality but also on how correlated reconstructions are composed; whether this effect transfers across regimes remains open.

Our contributions are:

1. On the 60-span assembly-held-out cohort, we provide evidence for cross-view fidelity selection: the unmodified whole-view score identifies the better aligned shared segment across independently inverted targets. Boundary outcomes, single-owner loss decomposition, and fixed-sample score reassignment distinguish useful score–view correspondence from a generic score distribution.
2. We realize the selection principle in FC–F and CP–F. On fixed reconstructed views, changing only the deterministic FC ownership rule adds 222 matched reference positions, a within-sample algorithmic consequence rather than a population or deployment causal effect. CP–F additionally addresses coordinate drift, but its small recovery gain over FC–F comes with more insertions and edits rather than uniform sequence improvement.

2 Background and Related Work

2.1 Dense retrieval and overlapping observations

Dense passage retrieval maps text and queries into a shared vector space [2]; RAG uses retrieved content as non-parametric context for generation [1]. GTR scales T5-based dual encoders and is the encoder family used in our experiment [3]. Before retrieval, the corpus must be divided into units that fit the encoder. Independent non-overlapping units can sever useful context, so an indexer may place windows of length c every $s < c$ tokens. Adjacent windows then share $c - s$ source tokens. In the threat considered here, exposure includes document grouping and window order as well as the chunk geometry; the attacker therefore observes correlated vectors whose generating strings have a known partial overlap.

This work concerns post-exposure reconstruction from those raw vectors; compromising the vector store and extracting passages through a RAG interface are outside its scope.

2.2 Text embedding inversion

Song and Raghunathan analyze information leakage from embeddings [4], and GEIA reconstructs sentences using conditional generation [5]. Vec2Text repeatedly re-embeds reconstruction hypotheses and uses their agreement with a fixed target embedding during iterative correction [6]. Its analysis further reports a strong but imperfect association between target–hypothesis cosine similarity and reconstruction BLEU [6]. This supports similarity as a useful signal within one target's search, but does not establish that raw scores are calibrated or otherwise comparable across target embeddings. We reuse the score unchanged and test precisely that cross-view ownership decision.

Related work differs in access and training assumptions, cross-model generalization, language, and deployment context. Under alternative access and training assumptions, TEIA transfers through a surrogate without victim-model queries [8], Zero2Text performs per-instance online alignment instead of offline inversion training [9], and conditional masked-diffusion inversion trains encoder-specific denoisers [10]. For cross-model and data generalization, ALGEN aligns embedding spaces from few examples [11], while ZSInvert targets universal zero-shot inversion [12]. Multilingual work examines cross-lingual risk and defenses [13]. Across these axes, the primary reconstruction unit remains an individual target embedding. Our matched-setting experiment instead holds the inverter fixed and asks how correlated outputs from multiple independently inverted targets should be combined.

At the deployment axis, Zhuang et al. study how representation properties and transformations affect both Vec2Text recoverability and retrieval effectiveness [14]. We hold the encoder and inverter fixed and instead vary the composition of multiple reconstructions produced from overlapping source windows.

2.3 RAG extraction and our positioning

CopyBreakRAG, whose earlier version circulated as RAG-Thief, is a black-box application-level extraction attack that explicitly exploits overlap [15]. After eliciting a plaintext chunk through a RAG response, it uses visible character overlap at that chunk's beginning or end to construct anchor queries intended to retrieve neighboring chunks. Our observation channel and objective differ: the attacker starts from exposed raw chunk embeddings before their plaintext or source-token correspondence is known, independently inverts them into noisy views, and uses overlap to resolve reconstruction errors rather than to crawl the retriever. Our contribution is fidelity-guided composition of independently inverted overlapping embeddings.

Across these strands, the remaining gap is how to combine independently inverted embeddings when index construction deliberately repeats source content. Our unit of attack is an ordered set of raw embeddings linked by an attacker-known overlap policy. We test whether whole-view fidelity resolves complementary errors on shared content through two source-free realizations: FC–F on nominal coordinates and CP–F under coordinate drift.

3 Threat Model and Problem Formulation

3.1 System and attacker

Let the evaluated 136-token source span be a sequence of content-token IDs $d = (t_0, \ldots, t_{L-1})$. An indexer creates ordered windows

$$x_i = (t_{is}, \ldots, t_{is+c-1}), \tag{1}$$

where $c = 31$ and $s = 15$ in our primary configuration. Each window is independently encoded as $e_i = E(x_i)$. Adjacent source windows share 16 tokens. For $L = 136$, the overlap policy produces eight vectors; the comparison no-overlap policy produces five vectors spanning the same source tokens.

The geometry parameters are linked rather than independently chosen. The public checkpoint's 32-position interface fixes 31 content tokens plus EOS. The experiment then chooses stride 15 and an eight-window chain; nominal overlap follows as $31-15 = 16$, and covered span length follows as $31+7(15) = 136$. The records treat stride 15 and eight windows as protocol choices, not optimized values. Holding the resulting geometry fixed lets the ownership comparison change how the same reconstructed views are selected rather than changing their coverage.

The attacker obtains the ordered target embeddings, knows their nominal order, stride, window coordinates, and other chunk geometry, can run the matching Vec2Text inverter, and can query the target encoder to embed generated hypotheses. What remains unknown is the ground-truth source-token placement inside noisy reconstructions, hence the correct reconstructed cross-view alignment and owner. Table 1 makes this information boundary explicit: the attack uses vector-side information and generated hypotheses, while source truth and correctness labels remain evaluation-only. A vector-database snapshot or backup retaining document identifiers, chunk order, and geometry metadata is one plausible exposure. This is post-compromise analysis, not end-to-end RAG extraction: it neither obtains vectors through application queries nor bypasses access control. The evaluated instantiation requires embeddings, document grouping and order, attacker-known chunk geometry, compatible encoder query access, and a matching inverter.

3.2 Reconstruction objective

The fixed inverter produces one Top-1 reconstruction per target:

$$\hat{x}_i = I(e_i). \tag{2}$$

Assembly constructs $\hat{d}$ from these hypotheses, their recorded fidelity scores, and known geometry; source references enter only subsequent evaluation. The attack objective is to maximize exact recovery of the 136 protected source positions. Section 5.5 later defines source-aligned quantities used to diagnose ownership separately from the accuracy of an assembled output.

Table 1. Threat model. "Available" denotes information used by the attack at inference time.

Category	Assumption
Available	Exposed raw chunk embeddings with retained document grouping and order; target GTR encoder access for re-embedding hypotheses; frozen Vec2Text inverter/corrector; attacker-known chunk length, stride, and window coordinates; reconstructed Top-1 token sequences.
Unavailable	Raw source text; source-token positions; ground-truth correspondence between views; correctness labels; oracle view choices.
Out of scope	Compromising the vector store, bypassing authentication, extracting embeddings through a black-box RAG API, and defeating approximate-nearest-neighbor (ANN) compression, quantization, or added noise.

4 Cross-View Fidelity Selection

The contribution is the relative selection rule: use each original view's whole-view fidelity to choose an owner for aligned shared content. FC–F and CP–F instantiate that rule with nominal and progressively estimated correspondence, respectively. Figure 1 separates three stages. The top box fixes the reconstructed views, scores, order, and geometry. Inside the dashed attack boundary, FC–L and FC–F differ only in ownership on fixed coordinates. The constrained outputs are not a symmetric factorial pair: CP–L combines lower-index ownership with adjacent-pair boundary input, whereas CP–F combines fidelity ownership with accumulated-output boundary input and is therefore a whole-method contrast. Below the boundary, the source reference scores all four sequences but never informs assembly.

4.1 Independent inversion and embedding fidelity

Every overlap and no-overlap vector uses the repository's pre-existing strong-reproduction setting: 50 correction steps, sequence-beam width 8, and Top-1 output. It was fixed before final-cohort outcomes, is held constant across comparisons, and is not claimed optimal. Final strings are retokenized with the GTR tokenizer without special-token addition or truncation; EOS and padding (PAD) IDs are removed.

For each view, the implementation records a single reconstruction-level score

$$f_i = \cos(E(\hat{x}_i), e_i)\,, \tag{3}$$

where $E(\hat{x}_i)$ is the embedding of the final hypothesis and e_i is the protected target. This is the candidate's `local_score`: one consistency score for the en-

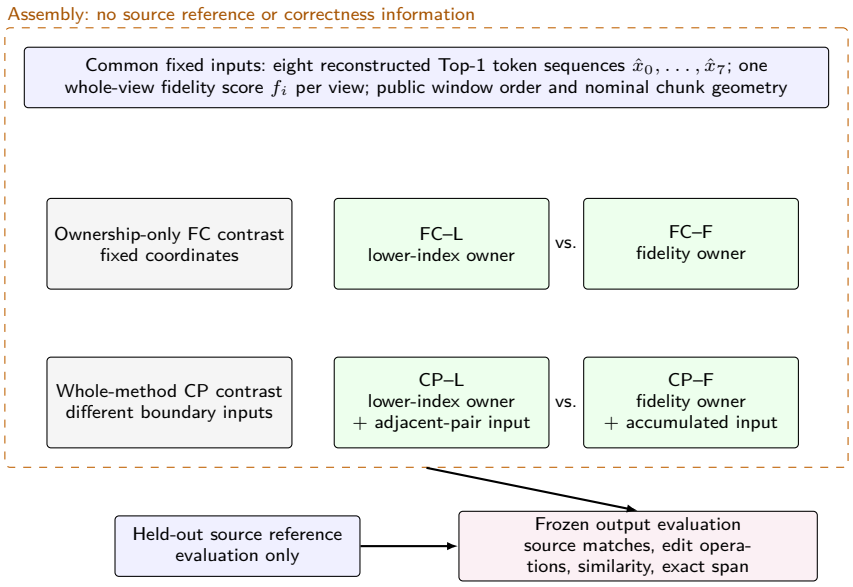


Fig. 1. Common inputs and the two non-equivalent output comparisons. The four outputs use the same fixed Top-1 reconstructions. FC denotes fixed-coordinate assembly: FC–L and FC–F differ only in lower-index versus corresponding-fidelity ownership. CP denotes constrained progressive assembly, but its two retained outputs do not form an ownership-only pair: CP–L uses lower-index ownership with adjacent original-pair input, whereas CP–F uses fidelity ownership with accumulated-output input and is therefore a whole-method contrast. Ground-truth source content enters only evaluation outside the dashed assembly boundary.

tire reconstructed view, rather than calibrated token confidence. We choose it because correction-based inversion already exposes the value at inference time and because it directly measures whether a generated view maps back toward the protected vector. The experiment therefore asks whether an existing signal transfers to cross-view selection without training a confidence model. Instrumentation verifies that the recorded candidate is exactly the public Top-1 hypothesis in token-ID space.

4.2 Fixed-coordinate realization

Fixed-coordinate assembly places each prediction in the nominal 31/15 layout and chooses the token from the covering view with higher fidelity. It is the cleanest ownership test because correspondence is supplied by known geometry rather than inferred from noisy strings. Predictions shorter than 31 positions contribute nulls, overlength positions are clipped, and ties and triple coverage

select the lowest-index view. This arbitrary source-free symmetry break is deterministic. The FC–L control replays the unchanged routine with tied scores, invoking that rule. FC–F versus FC–L therefore changes ownership alone while retaining clipping, null, length, and endpoint rules.

4.3 Constrained suffix–prefix alignment

Prediction lengths and boundaries can drift because inversion makes insertions and deletions. For each merge, let R_{i-1} be the accumulated sequence and let $\hat{x}_i$ be the next reconstructed view. We consider only suffix widths ℓ and prefix widths r in

$$\mathcal{W} = \{14, 15, 16, 17, 18\}, \tag{4}$$

clipped to the available prediction lengths. This ± 2 band and the weight below are frozen geometry-based development heuristics, not learned or optimized parameters. They permit small drift while excluding distant matches.

For each pair (ℓ, r), we compute unit-cost Levenshtein distance [16] between the suffix and prefix and minimize

$$J(\ell, r) = \mathrm{Lev}\big(\mathrm{suf}_\ell(R_{i-1}), \mathrm{pre}_r(\hat{x}_i)\big) + 0.5\big(|\ell - 16| + |r - 16|\big). \tag{5}$$

Match cost is zero; substitution and gap costs are one. Appendix A gives the deterministic tie-breaking order.

The first term rewards textual agreement, while each unit of total width deviation costs half an edit. With integer edit distance and smaller deviation winning objective ties, a one-sided one-token shift must save one edit, and shifting both widths by one must save two; saving only one in the latter case ties and retains the smaller deviation. Thus shorter comparisons win only with better candidate-only fit, and deterministic tie-breaking makes assembly repeatable.

4.4 Fidelity ownership and gaps

An aligned column contains (a, b), where either token may be null. If $a = b$, the token is emitted once. Otherwise one entire aligned shared segment is owned by the view with higher fidelity: the right view wins only if $f_i > f_{i-1} + 10^{-12}$. The 10^{-12} threshold is a numerical tolerance, not an effect threshold. Within it, the lower-index view supplies an arbitrary but deterministic source-free symmetry break. A substitution emits the owner's token. At a gap, the rule emits the owner's token when non-null and otherwise omits the column, preserving the whole-segment ownership intervention rather than switching owners only to retain a token.

With length 31 and stride 15, distance-two windows meet at isolated endpoints, yielding six source positions of multiplicity three in an eight-window span. At each such endpoint, known geometry designates the relevant first or last predicted boundary column and the deterministic rule keeps the lower-index view.

4.5 Constrained progressive realization

Starting from $R_0 = \hat{x}_0$, the algorithm repeats

$$R_i = \text{merge}(R_{i-1}, \hat{x}_i), \qquad i = 1, \ldots, 7. \tag{6}$$

The accumulated sequence supplies the left suffix. Shared-segment ownership uses f_{i-1}, the score of the immediately preceding original window, as a source-free proxy without token-level provenance. After resolving the aligned overlap, the unconsumed suffix of $\hat{x}_i$ is appended. No global path search or learned component is used.

The boundary search evaluates at most 25 pairs of short strings. The method is deterministic given the reconstructed Top-1 views and their fidelities; runtime and policy-cost accounting appear in Appendix B.

5 Experimental Setup

5.1 Data, encoder, and inverter

We use passages from the public Natural Questions benchmark [7] distributed through BEIR [17]. The encoder and tokenizer are GTR-T5-base at revision `9801579...`, with the public `jxm/gtr__nq__32` inverter and corrector. This matched public GTR/NQ checkpoint controls cross-model inversion failure so the experiment tests selection among viable reconstructions rather than checkpoint mismatch. Model inputs contain 31 content IDs followed by EOS, matching its 32-position regime. Vec2Text reports training the inverter on five million Wikipedia passages selected from NQ [6]; final-passage membership is unknown, so assembly-held-out does not imply checkpoint-training-disjoint. Full revisions are retained in the manifest.

Round-trip stability controls a representation confound before inversion. A token slice taken from a fully tokenized document can change when its raw substring is tokenized alone, especially at a boundary. Without this filter, an apparent assembly error could be caused by presenting a different token sequence to the inverter rather than by reconstruction or ownership. We therefore tokenize the full document once with offset mappings, extract the raw substring covering an intended 31-token whole-document slice, tokenize that substring independently without special tokens or truncation, and require exact equality with the intended content-ID slice. Results describe the resulting controlled subset of NQ documents rather than all NQ passages.

The dataset revision is pinned (`b7253e6c...e3b46`); the full identifier is retained in the manifest. We scan 1,000 records and retain documents that support a 136-source-token span whose eight 31-token windows pass the check. When several valid $K = 8$ chains exist, the constructor selects the first in ascending source-token order. This deterministic, result-blind rule avoids choosing a span by reconstruction quality; the records do not establish that earliest starts are otherwise preferable. Of 369 length-eligible documents, 210 satisfy the check; the other 159 (43.1%) have no all-window-stable chain. Exact tokenizer-ID deduplication removes no further documents.

5.2 Split discipline and freezing

The 210 eligible documents are deterministically ordered, with positions 0–149 treated as development-side and positions 150–209 reserved for final evaluation. Here, SHA positions mean zero-based ranks under that deterministic SHA-256 ordering, not substrings of a digest. The final 60 are the entire remainder after the 150 prior documents; the retained freeze record lists no unused reserve and records their inversion outputs and evaluation outcomes as unseen before the algorithm freeze. They are therefore assembly-held-out under the retained-record provenance. Table 2 records the historical roles and isolates this remainder in its last row.

Table 2. Deterministic cohort roles. "SHA positions" are zero-based ranks after deterministic SHA-256-key ordering. For the frozen assembly protocol, all 150 earlier documents are development-side; the final 60 were not used for assembly development, and their inversion outputs and evaluation outcomes were unavailable before the freeze.

Protocol role	**Historical use**	**SHA positions**	n
Development-side	Assembly design	0–29	30
Development-side	Fidelity validation	30–89	60
Development-side	Prior assembly access	90–149	60
Final test	Frozen validation	150–209	60

Manifest and integrity-audit details are reported in Appendix B.

5.3 Chunk policies and actual-output baselines

All output comparisons cover the same 136 source positions. We retain four sequences built from the same eight cached Top-1 overlap views. The key mechanistic comparison is FC–F versus FC–L: both use identical fixed coordinates, clipping, null handling, and tie rules, but FC–F assigns the aligned shared segment to the higher-fidelity view while FC–L assigns it to the lower-index view. Their within-sample algorithmic difference isolates the effect of the owner rule on these fixed reconstructed views and this fixed-coordinate pipeline.

FC–F versus CP–F compares nominal-coordinate and progressively estimated correspondence under fidelity ownership. CP–F versus CP–L is instead a whole-method contrast: CP–L uses lower-index ownership and adjacent original hypotheses for boundary input, whereas CP–F uses fidelity ownership and accumulated output.

Finally, the prespecified complete-policy comparison asks what is recovered when the index does or does not deliberately repeat source positions. The overlap policy exposes eight vectors at starts $0, 15, \ldots, 105$. Operationally, the capacity-safe no-overlap constructor starts at the leftmost uncovered source position, shortens the proposed raw substring until its standalone tokenization fits 31

content IDs, emits that window, and advances by the source positions it covered. It therefore covers each of the 136 positions once in five windows; 42 of 300 final-cohort windows require one right-edge deferral. The comparison changes vector count and coverage pattern with the policy, so it is not an equal-budget estimate of overlap alone.

5.4 Metrics and statistical analysis

The predeclared primary metric is *source-token recovery*: the number of exact matches to protected reference positions. We align the final predicted sequence to the 136-token reference with a deterministic unit-cost edit-distance traceback and count matched reference positions, even when reconstruction shifts length. Insertions cannot reveal an additional protected reference position, so they are excluded from recovery and reported separately. This metric is distinct from Token F1, BLEU, and ROUGE-L, which can reward unordered overlap or fluent similarity without identifying recovered reference positions. Appendix A records the traceback rule and sensitivity results.

Because recovery alone does not penalize extra output, secondary metrics report sequence quality and error cost: normalized sequence similarity,

$$\max\left(0, 1 - \frac{\mathrm{Lev}(\hat{d}, d)}{|d|}\right), \tag{7}$$

total edit distance, substitution, insertion and deletion counts, and mean signed output-length bias (prediction length minus 136). Span exact match is tertiary and descriptive: one error makes an otherwise accurate 136-token span fail, and separately tokenizing raw chunk strings can change subword IDs at a concatenation boundary. The perfect-input representation controls appear in Appendix C.

Unless stated otherwise, bootstrap intervals resample the 60 passage IDs with replacement. They are pointwise and unadjusted, quantify empirical sensitivity to the composition of this fixed passage cohort, and are neither population-generalization intervals nor article-clustered intervals because original-article linkage is unavailable.

The frozen final comparison was CP–F versus no-overlap; CP–F versus CP–L and FC–F were secondary protocol comparisons. The boundary, FC–L, and reassignment analyses were developed later as post-hoc exploratory mechanism analyses on the same 60-span cohort. They provide within-sample evidence for how cross-view selection operates and motivate prospective replication rather than converting this cohort into an independent confirmation. Appendix A distinguishes interval procedures and the remaining sensitivity analyses.

5.5 Shared-source selection diagnostics

For each adjacent original-view pair, the shared-source analysis retains positions covered by exactly two views. The nominal source-window order and coordinates

are known to the attacker, but the ground-truth mapping from noisy reconstructed tokens back to those source positions—and therefore their correctness—is not. Per-view evaluation alignment supplies that mapping only after inference. The *source-aligned fidelity selector* chooses the token from the covering view with higher fidelity. The *restricted selection oracle* instead chooses by correctness between the two adjacent views while retaining the lower-index view at triple-coverage endpoints. Both are ground-truth-assisted diagnostic references, not attack outputs.

The fidelity owner is right only when its score exceeds the left score by more than 10^{-12}, with ties assigned left. We report conditional position success where exactly one view is correct, boundary success where aggregate shared-source quality differs, and selection regret relative to the better single owner. The final cohort contains 6,000 multiplicity-two (M2) positions, distinct from the 8,160 positions used for assembled-output recovery.

6 Evaluation

6.1 Whole-view fidelity selects shared-source quality

The first result supports the cross-view selection signal. The shared-source mapping contains 6,000 M2 positions: 5,603 are correct in both views, 371 in exactly one view, and 26 in neither. The exactly-one-correct subset is informative because ownership can change its outcome. Corresponding fidelity selects 347 of these 371 positions (93.53%). The pointwise bootstrap interval is [87.47%, 98.68%]. Thus, when independently reconstructed views disagree in correctness, the score attached to the original view usually points toward the recoverable token. These positions are nested within passages rather than independent trials.

The same pattern holds at the decision unit used by assembly. Across 420 original adjacent boundaries, 80 have unequal shared-source quality and occur in 36 passages. Fidelity selects the better owner at 76/80 boundaries (95.0%; pointwise passage-bootstrap interval [90.36%,98.80%]). The 24 conditional position failures decompose into seven losses unavoidable under any single-boundary owner and 17 units of regret from choosing the worse owner; positive regret is concentrated at four boundaries in four passages. Selection regret therefore accounts for most of these 24 failures, despite occurring at few boundaries.

Figure 2 shows the underlying score–quality relation rather than only its success count. For boundary b, $\Delta f_b = f_L - f_R$ compares whole-view fidelities and $\Delta q_b = q_L - q_R$ compares the numbers of correct shared-source positions under evaluation-only alignment. The 76 correctly ordered informative boundaries have matching nonzero signs; the four errors have opposing signs.

This diagnostic matches FC's ownership granularity: one owner governs the whole nominal overlap. It does not reproduce CP's inferred alignment or selected boundary strings. The FC–F versus FC–L contrast below provides the direct link from ownership to assembled output.

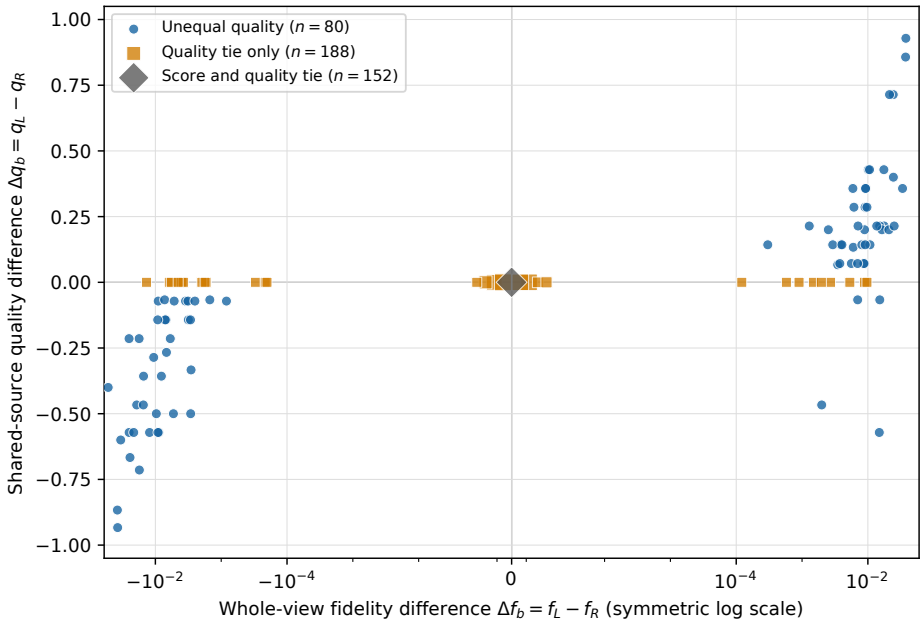


Fig. 2. Whole-view fidelity difference versus shared-source quality difference for all 420 adjacent boundaries. The horizontal axis is $\Delta f_b = f_L - f_R$ (symmetric-log scale), and the vertical axis is $\Delta q_b = q_L - q_R$; positive values favor the left view and negative values favor the right. Among the 80 unequal-quality boundaries (circles), quadrants I and III indicate correct sign agreement, containing 76/80. Squares mark 188 quality ties with unequal scores; diamonds mark 152 joint score–quality ties. Marker area encodes the number of boundaries at the same plotted coordinate and in the same category. Shapes, zero lines, and quadrant semantics preserve the comparison in grayscale.

6.2 Correct score–view correspondence matters

Table 3 summarizes source-aligned ownership diagnostics and the three fixed-sample reassignment stress tests.

The first three rows check against always favoring one side. All rows keep the Top-1 outputs, source mapping, and evaluation boundaries fixed; only ownership or score assignment changes. Corresponding fidelity recovers 5,950 shared positions, compared with 5,763 for always-left and 5,814 for always-right. Its paired mean passage gain is +3.12 points over always-left (pointwise 95% bootstrap interval [+2.07,+4.25]) and +2.27 over always-right ([+1.33,+3.33]). These sample contrasts are inconsistent with a simple left/right preference but are not confirmatory tests.

The last three rows summarize 1,000 fixed-sample assignments per control; corresponding scores recover 99.17%. Between-passage scores average 96.30% recovery, and their 2.5–97.5% assignment quantiles span 95.43–97.13%. Within-passage scores average 96.38%, with quantiles spanning 95.50–97.27%; random

Table 3. Shared-source selection on multiplicity-two positions (post hoc). Correct (/6,000) counts selected correct tokens, including both-correct positions. Conditional P_{pos} is the percentage selecting the correct view among 371 exactly-one-correct positions. Boundary success is the percentage selecting the higher-quality owner among 80 unequal-quality boundaries. Regret positions count uniquely recoverable positions lost relative to the better single owner. The last three rows are fixed-sample stress-test means from 1,000 assignments per control, with outputs and mappings fixed.

Selection rule	Correct /6,000	P_{pos} /371 (%)	Boundary /80 (%)	Regret positions
Corresponding fidelity	5,950	93.53	95.00	17
Always left	5,763	43.13	48.75	204
Always right	5,814	56.87	51.25	153
Random boundary owner	5,787.42	49.71	49.73	179.58
Between-passage scores	5,778.08	47.19	49.41	188.92
Within-passage scores	5,782.59	48.41	50.14	184.41

ownership averages 96.46%, with quantiles spanning 95.65–97.28%. Corresponding scores outperform every recorded reassignment. This indicates that score–reconstruction correspondence matters on this fixed sample. The ranges are neither population confidence intervals nor formal permutation tests.

Under alternative minimum-edit mappings, the aggregate advantage remains positive: fidelity minus always-right is [+126,+140] matches, and fidelity minus always-left is [+161,+194]. These bounds concern ambiguity in the M2 source-aligned diagnostic mapping, not the traceback of an assembled output; Appendix C defines the bounds and their interpretation.

6.3 Ownership effect on source-free assembled output

Table 4 reports the four assembled sequences used to separate the ownership-only FC contrast from the whole-method CP contrast.

FC–F adds 222 matched reference positions (+2.72 points) over FC–L and reduces total edits by 278. The pointwise bootstrap interval is [+1.80,+3.74], with 31 wins, 29 ties, and no losses in passage-level recovery. Because the routine, inputs, coordinates, and length handling are identical, this within-sample algorithmic contrast estimates the effect of the owner rule on these fixed reconstructed views; it is not a population causal effect.

CP–F exceeds CP–L by 200 matches (+2.4510 points) and has 243 fewer edits. The prespecified pointwise passage-bootstrap interval is [+1.5809,+3.3946]. This supports the evaluated CP–F realization as a whole, but it does not isolate fidelity: CP–L uses adjacent original hypotheses, whereas CP–F uses accumulated output, so ownership and later boundary correspondence change together.

Table 4. Four assembled outputs on the same eight Top-1 views per span. FC: fixed-coordinate; CP–L: retained constrained adjacent-pair baseline; CP–F: accumulated-output constrained progressive assembly; L/F: lower-index/corresponding-fidelity ownership. The CP row also changes boundary inputs. FC–L is a post-hoc offline replay of the unchanged fixed-coordinate code with tied scores; the other outputs are preserved. M: matched reference positions; recovery is M/8,160 pooled reference positions; $S/I/D$: substitutions/insertions/deletions; E: total edits; Len.: total output tokens; Norm. sim.: span-mean normalized sequence similarity. Span-exact matches are counts out of 60.

Cell	M	Rec. (%)	S	I	D	E	Len.	Norm. sim. (%)	Span exact (/60)
FC–L	7,823	95.87	243	78	94	415	8,144	94.91	26
FC–F	8,045	98.59	79	22	36	137	8,146	98.32	38
CP–L	7,861	96.34	228	84	71	383	8,173	95.31	26
CP–F	8,061	98.79	70	41	29	140	8,172	98.28	35

6.4 Progressive tradeoff and evaluation sensitivity

Comparing CP–F directly with FC–F, progressive correspondence recovers 8,061 versus 8,045 reference positions (98.7868% versus 98.5907%). The 16 additional matches (+0.1961 points; 95% interval [+0.0613,+0.3431]) accompany nine fewer substitutions and seven fewer deletions, but 19 more insertions, three more total edits, and 26 more output tokens. Normalized sequence similarity is 98.2843% versus 98.3211%, and span-exact matches are 35 versus 38. Progressive correspondence therefore produces a small recovery gain, not a uniform sequence-quality improvement.

The deterministic traceback gives the 16-match point difference. A later local replay gives a CP–F minus FC–F sensitivity envelope of [+6,+23] matches, preserving the recovery ordering across retained minimum-cost alignments. This assembled-sequence traceback analysis is distinct from the M2 source-aligned mapping bounds above; Appendix A defines the range and its reproducibility boundary.

6.5 Prespecified Complete-Policy Comparison

The complete eight-vector progressive overlap policy recovers 8,061/8,160 reference positions (98.7868%) versus 7,911/8,160 (96.9485%) for the five-vector no-overlap policy. The 150-match gain is +1.8382 points (prespecified pointwise passage-bootstrap interval [+0.6250,+3.2353]; 28/25/7 passage-level recovery wins/ties/losses). Normalized sequence similarity rises from 95.6618% to 98.2843%, and total edits fall from 354 to 140. This policy-level result complements the mechanistic ownership finding but does not isolate overlap: both arms use the overlap-selected cohort, while observation count, coverage pattern, and raw-boundary representation differ. A fixed-budget estimate would require equalizing those inputs and independently addressing shared eligibility. The original table, paired comparison, and edit breakdown appear in Appendix B.

7 Discussion and Limitations

Main implication. An ordered overlapping index should not be assessed as a bag of independent embeddings. In the evaluated setting, each reconstruction's whole-view fidelity carries relative information about which view better preserves their shared content. The 76/80 boundary result shows the signal at the method's decision unit, score reassignment links the gain to the correct view, and the 222-match FC–F improvement shows that the signal changes assembled-output recovery. Together, these results extend fidelity from ranking candidates for one target to selecting between independently inverted targets that repeat source content.

What progressive correspondence adds. The mechanism does not require progressive alignment to be useful: fixed-coordinate ownership supplies the cleanest evidence. Progressive assembly addresses a different problem—boundary drift after insertions and deletions—and adds only 16 recovered positions while worsening insertions, total edits, normalized sequence similarity, and span-exact matches. Its value is therefore metric-dependent; the appendix separately reports assembled-sequence traceback and source-aligned mapping sensitivity checks.

Evaluated scope. The evidence comes from one GTR-T5-base/Vec2Text pairing on English NQ passages with 31 content tokens plus EOS, stride 15, nominal overlap 16, 50 correction steps, and sequence-beam width 8. The overlap-defined round-trip-stability rule retains 210 of 369 length-eligible documents, so the result applies to this controlled subset. Independent corpora, encoders, languages, geometries, and tokenizer-unstable documents remain replication settings.

Policy interpretation. The eight-versus-five comparison measures the confidentiality difference between complete chunking policies. Because observation count, coverage pattern, and raw-boundary representation change together, it is not an equal-budget causal estimate of overlap alone. That distinction does not affect the controlled FC–F/FC–L ownership result, which uses the same eight reconstructed views.

Security use and mitigations. For an exposed ordered raw-vector index, risk assessment based only on individual inversion quality can miss the effects of choosing among imperfect correlated views. Protecting embeddings together with their grouping and order is therefore relevant to the stated post-exposure threat model. Compressed, perturbed, unordered, encrypted, or query-free settings and their retrieval tradeoffs require separate evaluation [14].

8 Conclusion

Whole-view reconstruction fidelity can help choose the more accurate view of a shared source segment. In this matched GTR/NQ sample, the correct score–view

correspondence matters, and the FC–F realization adds 222 matched reference positions to FC output. The CP–F realization addresses coordinate drift but offers only a limited recovery–edit tradeoff over FC–F. These selection findings complement the prespecified complete-policy comparison. Thus, cross-view fidelity selection improves reconstruction in this overlap-stable matched cohort; whether either realization generalizes across encoders, corpora, chunk geometries, and deployment conditions remains open.

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A Reproducibility Details

Boundary selection and assembly traceback. For each candidate (ℓ, r), match cost is zero and substitution and gap costs are one. Candidate boundaries are ordered lexicographically by J, total width deviation, raw edit distance, absolute right-width deviation, absolute left-width deviation, longer right width, and then smaller left width. The selected boundary strings use a deterministic dynamic-programming traceback that prefers an exact-match diagonal, a substitution diagonal, a left-token-to-gap step, and then a gap-to-right-token step.

Evaluation alignment. Per-view diagnostics align each reconstructed view to its own 31-token reference window and then offset local reference positions into span coordinates. Final-output evaluation instead aligns one assembled sequence to the full 136-token reference. Both use unit-cost edit distance and a deterministic traceback that prefers an exact diagonal, a substitution diagonal, a step consuming only the prediction, and then a step consuming only the reference. Substitutions map to incorrect reference positions; deletions are unrecovered; insertions create no reference position; and repeated token IDs follow this ordered trace rather than string containment. The per-view source coordinates are evaluation-only and are not passed to assembly.

A post-hoc local replay enumerates exact-match extrema across all minimum-cost alignments for each fixed output (Table 5). The resulting difference envelopes are [+195,+232] matches for FC–F minus FC–L, [+182,+207] for CP–F minus CP–L, and [+6,+23] for CP–F minus FC–F. These are evaluation-sensitivity ranges rather than sampling intervals, and their endpoints need not arise under one common tie-breaking policy.

Table 5. Post-hoc exact-match extrema from a local replay over all minimum-cost token-ID alignments on the fixed 60-span sample. These ranges describe alignment ambiguity, not sampling uncertainty. The prepared numerical audit package omits the replayed final sequences and cannot reproduce this table independently.

Method	Frozen matches	Optimal range
Fixed-coordinate lower-owner	7,823	7,823–7,850
No-overlap	7,911	7,911–7,930
Constrained fixed-owner	7,861	7,861–7,879
Fixed-coordinate fidelity	8,045	8,045–8,055
CP–F	8,061	8,061–8,068

Bootstrap and reassignment. The retained conditional-position procedure uses 10,000 paired passage resamples and seed 20260910. The later boundary, regret, and fixed-owner summaries use 10,000 paired passage resamples and seed 20260915. All reported intervals are pointwise without multiplicity adjustment. For conditional ratios, each draw resamples passage IDs, sums the selected passages' numerators and denominators, and recomputes the ratio; a zero denominator is recorded as NA. The cohort has one span for each of 60 distinct NQ passage IDs. Original-article linkage is unavailable, so these are passage-level intervals; tokens and boundaries are not independently resampled.

The score diagnostics use 1,000 assignments and seed 20260916. Random ownership draws one fair choice per boundary. Between-passage reassignment permutes complete eight-score vectors while preserving window index; within-passage reassignment permutes the original eight scores inside each passage. Self assignments are allowed, all assignments are recorded, and scores remain within

the same dataset, split, and geometry. Assignment quantiles describe variation on these fixed passages rather than population uncertainty or a formal permutation test.

B Experimental Provenance and Cost

The retained manifests order the 210 eligible documents by a deterministic SHA-256 key derived from the pinned dataset revision, seed, maximum window count, and passage ID. They record every document at positions 0–149 as development-side: 30 informed assembly design, 60 evaluated the fidelity mechanism, and 60 had been accessed in an earlier assembly study. They further record that the method and evaluation protocol were frozen before inversion outputs or evaluation outcomes for positions 150–209 became available. This is retained-record provenance rather than independent timestamp certification.

The frozen manifest records the algorithm, evaluation protocol, model and tokenizer revisions, dependency versions, input artifacts, and per-file hashes. A later local collection run rehashed and matched all 43 unique files referenced by those records. This verifies their identity at collection time, not freeze timing. The experimental snapshot was not a Git worktree and has no commit identifier; the current submission also has no public artifact URL. These details concern artifact provenance and leave the frozen numerical evaluation unchanged.

Three reproducibility tiers. First, the prepared numerical audit package recomputes reported aggregate differences and bootstrap intervals from projected per-span numerical records. Second, retained local artifacts support assembly replay, traceback envelopes, mapping bounds, and oracle cross-tabs. Third, end-to-end inversion and assembly require the source and reconstructed text, token IDs, embeddings, caches, and implementation code retained outside that package. Accordingly, numerical recalculation, local replay, and full experiment reproduction are distinct evidence levels.

Assembly required 0.421 seconds for all 60 final spans (0.007 seconds per span); inversion dominates attack cost. Table 6 reports post-hoc logical-policy accounting from recorded per-view search times. Cache sharing reduced 780 logical views to 720 unique inversions, and a separate instrumentation timer totals 22,526 seconds for those unique inversions.

Table 6. Post-hoc logical-policy cost accounting from stored telemetry. Shared views are charged to each policy; these are summed recorded search times, not independently timed isolated runs. Model loading and batching effects are excluded.

Policy	Views	Search seconds	Encoder calls	Hypothesis rows
Overlap	480	15,008.999	24,480	1,509,600
No-overlap	300	9,390.363	15,300	943,500

The following retained tables report the prespecified complete-policy comparison, secondary contrasts, and the pre-freeze representation control. Their definitions and calculation procedures were prespecified; numerical values and interval endpoints are observed results.

Table 7 separates assembled-output metrics from three source-aligned diagnostic references.

Table 7. Core final results (60 assembly-held-out 136-token spans; 8,160 protected reference positions). Recovery counts exact reference-position matches under the frozen alignment; the table reports normalized sequence similarity and the span exact match rate. Best assembled-output results are bold. Rows below the midrule are diagnostic references, not assembled sequences: they use source-aligned per-view mappings, so sequence metrics do not apply. The restricted oracle bounds only its two-view selection rule.

Method	Vec./ span	Rec. (%)	Norm. sim. (%)	Edits	Span exact (%)
No-overlap strong	5	96.9485	95.6618	354	18.33
CP–L: constrained fixed-owner	8	96.3358	95.3064	383	43.33
FC–F: fixed-coordinate fidelity	8	98.5907	**98.3211**	**137**	**63.33**
CP–F: progressive fidelity	8	**98.7868**	98.2843	140	58.33
Source-aligned fixed-owner [ref.]	8	96.5196	–	–	–
Source-aligned fidelity selector [ref.]	8	98.8113	–	–	–
Restricted selection oracle [bound]	8	99.1054	–	–	–

Table 8 reports paired passage-bootstrap differences and passage-level recovery win/tie/loss counts for each baseline.

Table 8. Paired span-level comparisons: CP–F minus each baseline. The comparison definitions and pointwise percentile-bootstrap procedure were prespecified (10,000 resamples; seed 20260910); numerical differences and interval endpoints are observed results. Win/tie/loss counts were added post hoc.

Baseline	Δ (pp)	95% bootstrap interval (pp)	W/T/L
No-overlap strong	+1.8382	[+0.6250, +3.2353]	28/25/7
Source-aligned fixed-owner	+2.2672	[+1.4828, +3.0882]	29/30/1
Constrained fixed-owner	+2.4510	[+1.5809, +3.3946]	30/30/0
Fixed-coordinate fidelity	+0.1961	[+0.0613, +0.3431]	11/47/2

Table 9 records the pre-freeze perfect-input representation control; it evaluates chunk reconstruction without inversion error.

Table 10 decomposes each assembled output into substitutions, insertions, deletions, and length bias.

Table 9. Perfect-input representation control on the pre-freeze 60-span mechanism cohort. Ground-truth chunks replace every inversion output; this is not a final-cohort attack result.

Representation	Recovery (%)	Seq. sim. (%)	Edits	Span exact (%)
No-overlap direct source IDs	100.0000	100.0000	0	100.00
No-overlap raw-text pipeline	99.7549	99.2402	62	28.33
Overlap window union/stitch	100.0000	100.0000	0	100.00

Table 10. Edit breakdown for sequence-producing methods. Length bias is mean prediction length minus 136.

Method	Sub.	Ins.	Del.	Length bias
No-overlap	186	105	63	+0.700
Constrained fixed-owner	228	84	71	+0.217
Fixed-coordinate fidelity	79	**22**	36	−0.233
CP–F	**70**	41	**29**	+0.200

C Supplementary Diagnostics and Post-hoc Audits

C.1 Source-aligned diagnostic references

The source-aligned fidelity selector recovers 8,063 reference positions (98.8113%), two more than the source-free assembled sequence under their respective evaluation mappings. Relative to source-aligned fixed-owner, the ratio of these aggregate gains is 98.93% (bootstrap interval [94.64%,102.48%]). This ratio compares different estimators rather than measuring a fraction of all recoverable content.

The restricted selection oracle reaches 99.1054% under source-aligned per-view correspondence, with the lower-index view fixed at triple-coverage endpoints. It measures headroom within that rule. The independently aligned sequence output and source-aligned references can be correct at different positions, so the oracle is not a bound on arbitrary sequence assembly.

C.2 Stored-record audits

All results in this subsection were computed after the primary analysis from stored final-cohort records; inversion and assembly were not rerun. Intervals from the prespecified evaluation procedure use $n = 60$, 10,000 passage resamples, seed 20260910, and the nearest sorted order statistic. They are pointwise intervals without multiplicity adjustment.

Table 8 gives post-hoc, per-passage comparisons of source-token recovery. The win, tie, and loss counts are respectively 28, 25, and 7 against no-overlap; 30, 30, and 0 against constrained fixed-owner; and 11, 47, and 2 against fixed-coordinate fidelity. The last comparison shows that the 16-position aggregate advantage is concentrated in a small minority of spans.

C.3 Mapping and historical controls

Outer-bound construction. For a fixed owner rule g, write the fidelity-minus-g match difference over M2 positions as $D_g = \sum_{d,i,p} w_{d,i,p} c_{d,i,p}$, where $c_{d,i,p}$ is correctness for one passage–view–source-position variable and $w_{d,i,p}$ is its signed owner coefficient under g. The same correctness variable is used wherever that view-position enters either method or boundary, and identical owner terms cancel before bounding. Forward/backward unit-edit dynamic programming identifies whether each reference position can be correct on any minimum-cost path, giving variable-wise limits $l_{d,i,p}$ and $u_{d,i,p}$. Relaxing the remaining within-view path dependence gives

$$\sum_{d,i,p} \min\{w_{d,i,p} l_{d,i,p}, w_{d,i,p} u_{d,i,p}\} \le D_g \le \sum_{d,i,p} \max\{w_{d,i,p} l_{d,i,p}, w_{d,i,p} u_{d,i,p}\}. \quad (8)$$

Every consistent minimum-edit mapping assigns each shared variable within these limits, so its aggregate difference lies inside this interval. The relaxation may combine variable extrema that cannot coexist on one traceback; the endpoints are therefore conservative outer bounds rather than attainable joint extrema.

Table 11 reports the resulting conservative M2 match-difference bounds for the fixed owner controls.

Table 11. Post-hoc per-view mapping sensitivity over 6,000 M2 positions. Differences are fidelity minus fixed-owner exact source matches. The outer bounds relax optimal-path dependence; their endpoints are not guaranteed to be jointly attainable. These bounds are not confidence intervals.

Fixed owner	Frozen mapping	Outer bounds
Always-right	+136	[+126,+140]
Always-left	+187	[+161,+194]

Among 55 ambiguous view–reference-position variables, 47 occur at M2 positions; after owner cancellation, 14 retain nonzero coefficients against always-right and 33 against always-left. Both lower endpoints are positive, so every feasible mapping covered by the relaxation preserves the aggregate M2 advantage in this sample. These linear bounds are not confidence intervals or exact joint ranges for boundary success, conditional position success, or regret. The 17 regret units equal 4.58% of the 371 uniquely recoverable opportunities, 4.67% of the best single-owner unique-position total, and 0.28% of all 6,000 M2 positions.

Table 12 retains the earlier cyclic-document and distance-two assignments; the corresponding-minus-control rows are explicitly post hoc and complement the repeated reassignment analysis in the main text.

Table 13 cross-tabulates oracle and final-sequence correctness.

The unrestricted post-hoc check finds a correct token in at least one covering view at 8,095 positions (99.2034%), compared with 8,087 for the frozen

Table 12. Source-aligned mechanism controls on the final cohort. The first three comparison definitions and procedures were prespecified; the two direct paired contrasts are explicitly post hoc. Numerical differences and interval endpoints are observed results. Intervals are pointwise passage-bootstrap intervals.

Comparison	**Δ (pp)**	**95% passage-bootstrap interval (pp)**
Corresponding per-view fidelity	+2.2917	[+1.5196, +3.1005]
Shuffled-document fidelity	−0.1961	[−0.9314, +0.4902]
Distance-two-view fidelity	+0.1593	[−0.5392, +0.8578]
Corresponding − shuffled *[post hoc]*	+2.4877	[+1.5196, +3.6520]
Corresponding − distance-two *[post hoc]*	+2.1324	[+1.1887, +3.2230]

Table 13. Post-hoc cross-tabulation of source-aligned oracle status and CP–F output status over 8,160 reference positions. Entries are counts. The unrestricted any-covering diagnostic was added after the primary analysis.

Oracle definition	**Both correct**	**Oracle only**	**CP–F only**	**Both wrong**
Frozen restricted selection	8,057	30	4	69
Post-hoc any covering	8,061	34	0	65

restricted oracle. The eight-position difference comes from the oracle's lower-index rule at triple coverage. Of the 65 positions where every covering view is wrong, 56 receive the residual label "all covering views wrong" and nine receive the higher-priority evaluation-ambiguity label. These cross-tabulations compare source-aligned oracle correctness with final-sequence correctness.

As an exploratory representation check within the same overlap-selected cohort, the policy difference is +1.3889 points for the 18 spans with exact no-overlap perfect-input concatenation (95% interval [−0.8170, +4.0850]) and +2.0308 points for the other 42 spans ([+0.5427,+3.8165]). The split retains the original asymmetric eligibility rule and is diagnostic rather than independent validation.

An unchanged GPU-free post-hoc perfect-input check on the final cohort yields 99.7794% recovery, 99.2279% normalized sequence similarity, 63 edits, and 30.00% span exact match for no-overlap, versus 100% on all measures for overlap. The 0.2206-point perfect-input gap documents representation mismatch within the complete-policy comparison; it is not subtracted from the +1.8382-point attack estimate.

C.4 Residual taxonomy

Table 14. Post-hoc residual source-position taxonomy for the CP–F output under final-sequence alignment. These labels are not directly interchangeable with the restricted oracle's source-aligned per-view mapping. Attribution is diagnostic rather than causal; prediction insertions (41) are separate.

Category	**Count**
All covering views wrong	56
Fidelity wrong direction	25
Evaluation ambiguity	10
Gap-handling failure	7
Alignment failure	1
Global-assembly failure	0
Total residual source positions	99

Table 14 assigns the 99 unrecovered reference positions to the frozen post-hoc hierarchy; the CP–F output also has 41 prediction insertions. Source-aligned fidelity is correct at eight positions where the source-free assembled output is not; the reverse occurs at six, yielding a net two-position difference. Within this cohort, "all covering views wrong" is the largest assigned residual category. This descriptive taxonomy neither establishes a general bottleneck nor excludes benefits from global assembly in other settings.